

WORLDS OF

SEPTEMBER 1962 • 35c



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CULTURAL EXCHANGE BY KEITH LAUMER
GORDON DICKSON • VANCE AANDAHL

THE SNOWBANK ORBIT

DRAMATIC ADVENTURE AT THE
ENDS OF THE SOLAR SYSTEM



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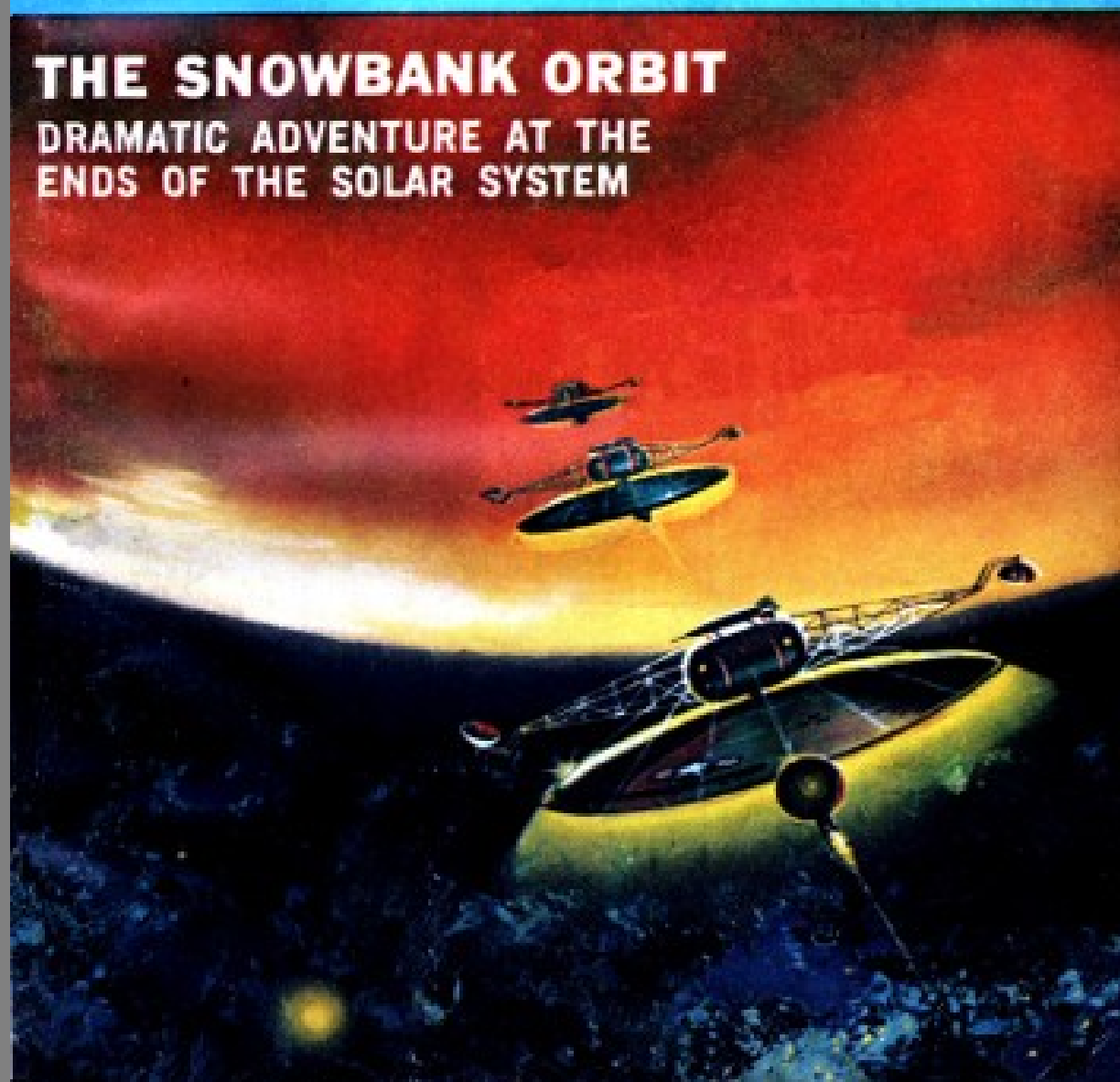


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*** START OF THE PROJECT GUTENBERG EBOOK THE
SNOWBANK ORBIT ***

THE SNOWBANK ORBIT

BY FRITZ LEIBER

**Earth could not stop the Enemy's
remorseless advance from outer
space. Neither could the Enemy!**

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The pole stars of the other planets cluster around Polaris and Octans, but Uranus spins on a snobbishly different axis between Aldebaran and Antares. The Bull is her coronet and the Scorpion her footstool. Dear blowzy old bitch-planet, swollen and pale and cold, mad with your Shakespearean moons, white-mottled as death from Venerean Plague, spinning on your side like a poisoned pregnant cockroach, rolling around the sun like a fat drunken floozie with green hair rolling on the black floor of an infinite bar-room, what a sweet last view of the Solar System you are for a cleancut young spaceman....

Grunfeld chopped off that train of thought short. He was young and the First Interstellar War had snatched him up and now it was going to pitch him and twenty other Joes out of the System on a fast curve breaking around Uranus—and so what! He shivered to get a little heat and then applied himself to the occulted star he was tracking through *Prospero's* bridge telescope. The star was a twentieth planetary diameter into Uranus, the crosslines showed—a glint almost lost in pale green. That meant its light was bulleting 1600 miles deep through the seventh planet's thick hydrogen atmosphere, unless he were seeing the star on a mirage trajectory—and at least its depth agreed with the time since rim contact.

At 2000 miles he lost it. That should mean 2000 miles plus of hydrogen soup above the methane ocean, an America-wide layer of gaseous gunk for the captain to play the mad hero in with the fleet.

Grunfeld didn't think the captain wanted to play the mad hero. The captain hadn't gone space-simple in any obvious way like Croker and Ness. And he wasn't, like Jackson, a telepathy-racked visionary entranced by the Enemy. Worry and responsibility had turned the captain's face into a skull which floated in Grunfeld's imagination when he wasn't actually seeing it, but the tired eyes deep-sunk in the dark sockets were still cool and perhaps sane. But because of the worry the captain always wanted to have the last bit of fact bearing on the least likely maneuver, and two pieces of evidence were better than one. Grunfeld found the next sizable star due to occult. Five-six minutes to rim contact. He floated back a foot from the telescope, stretching out his thin body in the plane of the ecliptic—strange how he automatically

assumed that orientation in free fall! He blinked and blinked, then rested his eyes on the same planet he'd been straining them on.

The pale greenish bulk of Uranus was centered in the big bridge spaceshield against the black velvet dark and bayonet-bright stars, a water-splotched and faded chartreuse tennis ball on the diamond-spiked bed of night. At eight million miles she looked half the width of Luna seen from Earth. Her whitish equatorial bands went from bottom to top, where, Grunfeld knew, they were spinning out of sight at three miles a second—a gelid waterfall that he imagined tugging at him with ghostly green gangrenous fingers and pulling him over into a hydrogen Niagara.

Half as wide as Luna. But in a day she'd overflow the port as they whipped past her on a near miss and in another day she'd be as small as this again, but behind them, sunward, having altered their outward course by some small and as yet unpredictable angle, but no more able to slow *Prospero* and her sister ships or turn them back at their 100 miles a second than the fleet's solar jets could operate at this chilly distance from Sol. G'by, fleet. G'by, C.C.Y. spaceman.

Grunfeld looked for the pale planet's moons. Miranda and Umbriel were too tiny to make disks, but he distinguished Ariel four diameters above the planet and Oberon a dozen below. Spectral sequins. If the fleet were going to get a radio signal from any of them, it would have to be Titania, occulted now by the planet and the noisy natural static of her roiling hydrogen air and seething methane seas—but it had always been only a faint hope that there were survivors from the First Uranus Expedition.

Grunfeld relaxed his neck and let his gaze drift down across the curving star-bordered forward edge of *Prospero's* huge mirror and the thin jutting beams of the port lattice arm to the dim red-lit gages below the spaceshield.

Forward Skin Temperature seven degrees Kelvin. Almost low enough for helium to crawl, if you had some helium. *Prospero's* insulation, originally designed to hold out solar heat, was doing a fair job in reverse.

Aft (sunward) Skin Temperature 75 degrees Kelvin. Close to that of Uranus' sun-lit face. Check.

Cabin Temperature 43 degrees Fahrenheit. Brr! The Captain was a miser with the chem fuel remaining. And rightly ... if it were right to drag out life as long as possible in the empty icebox beyond Uranus.

Gravities of Acceleration zero. Many other zeros.

The four telltales for the fleet unblinkingly glowed dimmest blue—one each for *Caliban*, *Snug*, *Moth*, and *Starveling*, following *Prospero* in line astern on slave automatic—though for months inertia had done all five ships' piloting. Once the buttons had been green, but they'd wiped that color off the boards because of the Enemy.

The gages still showed their last maximums. Skin 793 Kelvin, Cabin 144 Fahrenheit, Gravs 3.2. All of them hit almost a year ago, when they'd been ace-ing past the sun. Grunfeld's gaze edged back to the five bulbous pressure suits, once more rigidly upright in their braced racks, that they'd been wearing during that stretch of acceleration inside the orbit of Mercury. He started. For a moment he'd thought he saw the dark-circled eyes of the captain peering between two of the bulging black suits. Nerves! The captain had to be in his cabin, readying alternate piloting programs for Copperhead.

Suddenly Grunfeld jerked his face back toward the spaceshield—so violently that his body began very slowly to spin in the opposite direction. This time he'd thought he saw the Enemy's green flashing near the margin of the planet—bright green, viridian, far vividder than that of Uranus herself. He drew himself to the telescope and feverishly studied the area. Nothing at all. Nerves again. If the Enemy were much nearer than a light-minute, Jackson would esp it and give warning. The next star was still three minutes from rim contact. Grunfeld's mind retreated to the circumstances that had brought *Prospero* (then only *Mercury One*) out here.

When the First Interstellar War erupted, the pioneer fleets of Earth's nations had barely pushed their explorations beyond the orbit of Saturn. Except for the vessels of the International Meteor Guard, spaceflight was still a military enterprise of America, Russia, England and the other mega-powers.

During the first months the advantage lay wholly with the slim black cruisers of the Enemy, who had an antigravity which allowed them to hover near planets without going into orbit; and a frightening degree of control over light itself. Indeed, their principal weapon was a tight beam of visible light, a dense photonic stiletto with an effective range of several Jupiter-diameters in vacuum. They also used visible light, in the green band, for communication as men use radio, sometimes broadcasting it and sometimes beaming it loosely in strange abstract pictures that seemed part of their language. Their gravity-immune ships moved by reaction to photonic jets the tightness of which rendered them invisible except near the sun, where they tended to ionize electronically dirty volumes of space. It was probably this effective invisibility, based on light-control, which allowed them to penetrate the Solar System as deep as Earth's orbit undetected, rather than any power of travel in time or sub-space, as was first assumed. Earthmen could only guess at the physical appearance of the Enemy, since no prisoners were taken on either side.

Despite his impressive maneuverability and armament, the Enemy was oddly timid about attacking live planets. He showed no fear of the big gas planets, in fact hovering very close to their turgid surfaces, as if having some way of fueling from them.

Near Terra the first tactic of the black cruisers, after destroying Lunostrovok and Circumluna, was to hover behind the moon, as though sharing its tide-lockedness—a circumstance that led to a sortie by Earth's Combined Fleet, England and Sweden excepted.

At the wholly disastrous Battle of the Far Side, which was visible in part to naked-eye viewers on Earth, the Combined Fleet was annihilated. No Enemy ship was captured, boarded, or seriously damaged—except for one which, apparently by a fluke, was struck by a fission-headed anti-missile and proceeded after the blast to "burn," meaning that it suffered a slow and puzzling disintegration, accompanied by a dazzling rainbow display of visible radiation. This was before the "stupidity" of the Enemy with regard

to small atomic missiles was noted, or their allergy to certain radio wave bands, and also before Terran telepaths began to claim cloudy contact with Enemy minds.

Following Far Side, the Enemy burst into activity, harrying Terran spacecraft as far as Mercury and Saturn, though still showing great caution in maneuver and making no direct attacks on planets. It was as if a race of heavily armed marine creatures should sink all ocean-going ships or drive them to harbor, but make no assaults beyond the shore line. For a full year Earth, though her groundside and satellite rocketyards were furiously busy, had no vehicle in deep space—with one exception.

At the onset of the War a fleet of five mobile bases of the U. S. Space Force were in Orbit to Mercury, where it was intended they take up satellite positions prior to the prospecting and mineral exploitation of the small sun-blasted planet. These five ships, each with a skeleton five-man crew, were essentially Ross-Smith space stations with a solar drive, assembled in space and intended solely for space-to-space flight inside Earth's orbit. A huge paraboloid mirror, its diameter four times the length of the ship's hull, superheated at its focus the hydrogen which was ejected as a plasma at high exhaust velocity. Each ship likewise mounted versatile radio-radar equipment on dual lattice arms and carried as ship's launch a two-man chemical fuel rocket adaptable as a fusion-headed torpedo.

After Far Side, this "tin can" fleet was ordered to bypass Mercury and, tacking on the sun, shape an orbit for Uranus, chiefly because that remote planet, making its 84-year circuit of Sol, was currently on the opposite side of the sun to the four inner planets and the two nearer gas giants Jupiter and Saturn. In the empty regions of space the relatively defenseless fleet might escape the attention of the Enemy.

However, while still accelerating into the sun for maximum boost, the fleet received information that two Enemy cruisers were in pursuit. The five ships cracked on all possible speed, drawing on the solar drive's high efficiency near the sun and expending all their hydrogen and most material capable of being vaporized, including some of the light-metal hydrogen

storage tanks—like an old steamer burning her cabin furniture and the cabins themselves to win a race. Gradually the curving course that would have taken years to reach the outer planet flattened into a hyperbola that would make the journey in 200 days.

In the asteroid belt the pursuing cruisers turned aside to join in the crucial Battle of the Trojans with Earth's largely new-built, more heavily and wisely armed Combined Fleet—a battle that proved to be only a prelude to the decisive Battle of Jupiter.

Meanwhile the five-ship fleet sped onward, its solar drive quite useless in this twilight region even if it could have scraped together the needed boilable ejectant mass to slow its flight. Weeks became months. The ships were renamed for the planet they were aimed at. At least the fleet's trajectory had been truly set.

Almost on collision course it neared Uranus, a mystery-cored ball of frigid gas 32,000 miles wide coasting through space across the fleet's course at a lazy four miles a second. At this time the fleet was traveling at 100 miles a second. Beyond Uranus lay only the interstellar night, into which the fleet would inevitably vanish....

Unless, Grunfeld told himself ... unless the fleet shed its velocity by ramming the gaseous bulk of Uranus. This idea of atmospheric braking on a grand scale had sounded possible at first suggestion, half a year ago—a little like a man falling off a mountain or from a plane and saving his life by dropping into a great thickness of feathery new-fallen snow.

Supposing her solar jet worked out here and she had the reaction mass, *Prospero* could have shed her present velocity in five hours, decelerating at a comfortable one G.

But allowing her 12,000 miles of straight-line travel through Uranus' frigid soupy atmosphere—and that might be dipping very close to the methane seas blanketing the planet's hypothetical mineral core—*Prospero* would have two minutes in which to shed her velocity.

Two minutes—at 150 Gs.

Men had stood 40 and 50 Gs for a fractional second.

But for two minutes.... Grunfeld told himself that the only surer way to die would be to run into a section of the Enemy fleet. According to one calculation the ship's skin would melt by heat of friction in 90 seconds, despite the low temperature of the abrading atmosphere.

The star Grunfeld had been waiting for touched the hazy rim of Uranus. He drifted back to the eyepiece and began to follow it in as the pale planet's hydrogen muted its diamond brilliance.

III

In the aft cabin, lank hairy-wristed Croker pinned another blanket around black Jackson as the latter shivered in his trance. Then Croker turned on a small light at the head of the hammock.

"Captain won't like that," plump pale Ness observed tranquilly from where he floated in womb position across the cabin. "Enemy can feel a candle of *our* light, captain says, ten million miles away." He rocked his elbows for warmth and his body wobbled in reaction like a polly-wog's.

"And Jackson hears the Enemy think ... and Heimdall hears the grass grow," Croker commented with a harsh manic laugh. "Isn't an Enemy for a billion miles, Ness." He launched aft from the hammock. "We haven't spotted their green since Saturn orbit. There's nowhere for them."

"There's the far side of Uranus," Ness pointed out. "That's less than ten million miles now. Eight. A bare day. They could be there."

"Yes, waiting to bushwack us as we whip past on our way to eternity," Croker chuckled as he crumpled up against the aft port, shedding momentum. "That's likely, isn't it, when they didn't have time for us back in the Belt?" He scowled at the tiny white sun, no bigger a disk than Venus, but still with one hundred times as much light as the full moon pouring from it—too much light to look at comfortably. He began to button the inner cover over the port.

"Don't do that," Ness objected without conviction. "There's not much heat in it but there's some." He hugged his elbows and shivered. "I don't remember being warm since Mars orbit."

"The sun gets on my nerves," Croker said. "It's like looking at an arc light through a pinhole. It's like a high, high jail light in a cold concrete yard. The stars are highlights on the barbed wire." He continued to button out the sun.

"You ever in jail?" Ness asked. Croker grinned.

With the tropism of a fish, Ness began to paddle toward the little light at the head of Jackson's hammock, flicking his hands from the wrists like flippers. "I got one thing against the sun," he said quietly. "It's blanketing out the radio. I'd like us to get one more message from Earth. We haven't tried rigging our mirror to catch radio waves. I'd like to hear how we won the battle of Jupiter."

"If we won it," Croker said.

"Our telescopes show no more green around Jove," Ness reminded him. "We counted 27 rainbows of Enemy cruisers 'burning.' Captain verified the count."

"Repeat: if we won it." Croker pushed off and drifted back toward the hammock. "If there was a real victory message they'd push it through, even if the sun's in the way and it takes three hours to catch us. People who win, shout."

Ness shrugged as he paddled. "One way or the other, we should be getting the news soon from Titania station," he said. "They'll have heard."

"If they're still alive and there ever was a Titania Station," Croker amended, backing air violently to stop himself as he neared the hammock. "Look, Ness, we know that the First Uranus Expedition arrived. At least they set off their flares. But that was three years before the War and we haven't any idea of what's happened to them since and if they ever managed to set up housekeeping on Titania—or Ariel or Oberon or even Miranda or Umbriel.

At least if they built a station that could raise Earth I haven't been told. Sure thing *Prospero* hasn't heard anything ... and we're getting close."

"I won't argue," Ness said. "Even if we raise 'em, it'll just be hello-goodbye with maybe time between for a battle report."

"And a football score and a short letter from home, ten seconds per man as the station fades." Croker frowned and added, "If Captain had cottoned to my idea, two of us at any rate could have got off this express train at Uranus."

"Tell me how," Ness asked drily.

"How? Why, one of the ship's launches. Replace the fusion-head with the cabin. Put all the chem fuel in the tanks instead of divvying it between the ship and the launch."

"I haven't got the brain for math Copperhead has, but I can subtract," Ness said, referring to *Prospero's* piloting robot. "Fully fueled, one of the launches has a max velocity change in free-fall of 30 miles per second. Use it all in braking and you've only taken 30 from 100. The launch is still going past Uranus and out of the system at 70 miles a second."

"You didn't hear all my idea," Croker said. "You put piggyback tanks on your launch and top them off with the fuel from the other four launches. Then you've 100 miles of braking *and* a maneuvering reserve. You only need to shed 90 miles, anyway. Ten miles a second's the close circum-Uranian velocity. Go into circum-Uranian orbit and wait for Titania to send their jeep to pick you up. Have to start the maneuver four hours this side of Uranus, though. Take that long at 1 G to shed it."

"Cute," Ness conceded. "Especially the jeep. But I'm glad just the same we've got 70 per cent of our chem fuel in our ships' tanks instead of the launches. We're on such a bull's eye course for Uranus—Copperhead really pulled a miracle plotting our orbit—that we may need a sideways shove to miss her. If we slapped into that cold hydrogen soup at our 100 mps—"

Croker shrugged. "We still could have dropped a couple of us," he said.

"Captain's got to look after the whole fleet," Ness said. "You're beginning to agitate, Croker, like you was Grunfeld—or the captain himself."

"But if Titania Station's alive, a couple of men dropped off would do the fleet some good. Stir Titania up to punch a message through to Earth and get a really high-speed retrieve-and-rescue ship started out after us. *If* we've won the War."

"But Titania Station's dead or never was, not to mention its jeep. And we've lost the Battle of Jupiter. You said so yourself," Ness asserted owlishly. "Captain's got to look after the whole fleet."

"Yeah, so he kills himself fretting and the rest of us die of old age in the outskirts of the Solar System. Join the Space Force and See the Stars! Ness, do you know how long it'd take us to reach the nearest star—except we aren't headed for her—at our 100 mps? Eight thousand years!"

"That's a lot of time to kill," Ness said. "Let's play chess."

Jackson sighed and they both looked quickly at the dark unlined face above the cocoon, but the lips did not flutter again, or the eyelids. Croker said, "Suppose he knows what the Enemy looks like?"

"I suppose," Ness said. "When he talks about them it's as if he was their interpreter. How about the chess?"

"Suits. Knight to King Bishop Three."

"Hmm. Knight to King Knight Two, Third Floor."

"Hey, I meant flat chess, not three-D," Croker objected.

"That thin old game? Why, I no sooner start to get the position really visualized in my head than the game's over."

"I don't want to start a game of three-D with Uranus only 18 hours away."

Jackson stirred in his hammock. His lips worked. "They...." he breathed. Croker and Ness instantly watched him. "They...."

"I wonder if he is really inside the Enemy's mind?" Ness said.

"He thinks he speaks for them," Croker replied and the next instant felt a warning touch on his arm and looked sideways and saw dark-circled eyes in a skull-angular face under a battered cap with a tarnished sunburst. Damn,

thought Croker, how does the captain always know when Jackson's going to talk?

"They are waiting for us on the other side of Uranus," Jackson breathed. His lips trembled into a smile and his voice grew a little louder, though his eyes stayed shut. "They're welcoming us, they're our brothers." The smile died. "But they know they got to kill us, they know we got to die."

The hammock with its tight-swathed form began to move past Croker and he snatched at it. The captain had pushed off from him for the hatch leading forward.

Grunfeld was losing the new star at 2200 miles into Uranus when he saw the two viridian flares flashing between it and the rim. Each flash was circled by a fleeting bright green ring, like a mist halo. He thought he'd be afraid when he saw that green again, but what he felt was a jolt of excitement that made him grin. With it came a touch on his shoulder. He thought, the captain always knows.

"Ambush," he said. "At least two cruisers."

He yielded the eyepiece to the captain. Even without the telescope he could see those incredibly brilliant green flickers. He asked himself if the Enemy was already gunning for the fleet through Uranus.

The blue telltales for *Caliban* and *Starveling* began to blink.

"They've seen it too," the captain said. He snatched up the mike and his next words rang through the *Prospero*.

"Rig ship for the snowbank orbit! Snowbank orbit with stinger! Mr. Grunfeld, raise the fleet."

Aft, Croker muttered, "Rig our shrouds, don't he mean? Rig shrouds and firecrackers mounted on Fourth of July rockets."

Ness said, "Cheer up. Even the longest strategic withdrawal in history has to end some time."

IV

Three quarters of a day later Grunfeld felt a spasm of futile fear and revolt as the pressure suit closed like a thick-fleshed carnivorous plant on his drugged and tired body. Relax, he told himself. Fine thing if you cooked up a fuss when even Croker didn't. He thought of forty things to re-check. Relax, he repeated—the work's over; all that matters is in Copperhead's memory tanks now, or will be as soon as the captain's suited up.

The suit held Grunfeld erect, his arms at his sides—the best attitude, except he was still facing forward, for taking high G, providing the ship herself didn't start to tumble. Only the cheekpieces and visor hadn't closed in on his face—translucent hand-thick petals as yet unfolded. He felt the delicate firm pressure of built-in fingertips monitoring his pulses and against his buttocks the cold smooth muzzles of the jet hypodermics that would feed him metronomic drugs during the high-G stretch and stimulants when they were in free-fall again. When.

He could swing his head and eyes just enough to make out the suits of Croker and Ness to either side of him and their profiles wavy through the jutting misty cheekpieces. Ahead to the left was Jackson—just the back of his suit, like a black snowman standing at attention, pale-olive-edged by the great glow of Uranus. And to the right the captain, his legs suited but his upper body still bent out to the side as he checked the monitor of his suit with its glowing blue button and the manual controls that would lie under his hands during the maneuver.



Beyond the captain was the spaceshield, the lower quarter of it still blackness and stars, but the upper three-quarters filled with the onrushing planet's pale mottled green that now had the dulled richness of watered silk. They were so close that the rim hardly showed curvature. The atmosphere must have a steep gradient, Grunfeld thought, or they'd already be feeling decel. That stuff ahead looked more like water than any kind of air. It bothered him that the captain was still half out of his suit.

There should be action and shouted commands, Grunfeld thought, to fill up these last tight-stretched minutes. Last orders to the fleet, port covers being cranked shut, someone doing a countdown on the firing of their torpedo. But the last message had gone to the fleet minutes ago. Its robot pilots were

set to follow *Prospero* and imitate, nothing else. And all the rest was up to Copperhead. Still....

Grunfeld wet his lips. "Captain," he said hesitantly. "Captain?"

"Thank you, Grunfeld." He caught the edge of the skull's answering grin. "We *are* beginning to hit hydrogen," the quiet voice went on. "Forward skin temperature's up to 9 K."

Beyond the friendly skull, a great patch of the rim of Uranus flared bright green. As if that final stimulus had been needed, Jackson began to talk dreamily from his suit.

"They're still welcoming us and grieving for us. I begin to get it a little more now. Their ship's one thing and they're another. Their ship is frightened to death of us. It hates us and the only thing it knows to do is to kill us. They can't stop it, they're even less than passengers...."

The captain was in his suit now. Grunfeld sensed a faint throbbing and felt a rush of cold air. The cabin refrigeration system had started up, carrying cabin heat to the lattice arms. Intended to protect them from solar heat, it would now do what it could against the heat of friction.

The straight edge of Uranus was getting hazier. Even the fainter stars shone through, spangling it. A bell jangled and the pale green segment narrowed as the steel meteor panels began to close in front of the spaceshield. Soon there was only a narrow vertical ribbon of green—*bright* green as it narrowed to a thread—then for a few seconds only blackness except for the dim red and blue beads and semi-circles, just beyond the captain, of the board. Then the muted interior cabin lights glowed on.

Jackson droned: "They and their ships come from very far away, from the edge. If this is the continuum, they come from the ... discontinuum, where they don't have stars but something else and where gravity is different. Their ships came from the edge on a gust of fear with the other ships, and our brothers came with it though they didn't want to...."

And now Grunfeld thought he began to feel it—the first faint thrill, less than a cobweb's tug, of *weight*.

The cabin wall moved sideways. Grunfeld's suit had begun to revolve slowly on a vertical axis.

For a moment he glimpsed Jackson's dark profile—all five suits were revolving in their framework. They locked into position when the men in them were facing aft. Now at least retinas wouldn't pull forward at high-G decel, or spines crush through thorax and abdomen.

The cabin air was cold on Grunfeld's forehead. And now he was sure he felt weight—maybe five pounds of it. Suddenly aft was *up*. It was as if he were lying on his back on the spaceshield.

A sudden snarling roar came through his suit from the beams bracing it. He lost weight, then regained it and a little more besides. He realized it was their torpedo taking off, to skim by Uranus in the top of the atmosphere and then curve inward the little their chem fuel would let them, homing toward the Enemy. He imaged its tiny red jet over the great gray-green glowing plain. Four more would be taking off from the other ships—the fleet's feeble sting. Like a bee's, just one, in dying.

The cheekpieces and foreheadpiece of Grunfeld's suit began to close on his face like layers of pliable ice.

Jackson called faintly, "Now I understand. Their ship—" His voice was cut off.

Grunfeld's ice-mask was tight shut. He felt a small surge of vigor as the suit took over his breathing and sent his lungs a gush of high-oxy air. Then came a tingling numbness as the suit field went on, adding an extra prop against decel to each molecule of his body.

But the weight was growing. He was on the moon now ... now on Mars ... now back on Earth....

The weight was stifling now, crushing—a hill of invisible sand. Grunfeld saw a black pillow hanging in the cabin above him aft. It had red fringe around it. It grew.

There was a whistling and shaking. Everything lurched torturingly, the ship's jets roared, everything recovered, or didn't.

The black pillow came down on him, crushing out sight, crushing out thought.

The universe was a black tingling, a limitless ache floating in a larger black infinity. Something drew back and there was a dry fiery wind on numb humps and ridges—the cabin air on his face, Grunfeld decided, then shivered and started at the thought that he was alive and in free-fall. His body didn't feel like a mass of internal hemorrhages. Or did it?

He spun slowly. It stopped. Dizziness? Or the suits revolving forward again? If they'd actually come through—

There was a creaking and cracking. The ship contracting after frictional heating?

There was a faint stink like ammonia and formaldehyde mixed. A few Uranian molecules forced past plates racked by turbulence?

He saw dim red specks. The board? Or last flickers from ruined retinas? A bell jangled. He waited, but he saw nothing. Blind? Or the meteor guard jammed? No wonder if it were. No wonder if the cabin lights were broken.

The hot air that had dried his sweaty face rushed down the front of his body. Needles of pain pierced him as he slumped forward out of the top of his opening suit.

Then he saw the horizontal band of stars outlining the top of the spaceshield and below it the great field of inky black, barely convex upward, *that must, he realized, be the dark side of Uranus.*

Pain ignored, Grunfeld pushed himself forward out of his suit and pulled himself past the captain's to the spaceshield.

The view stayed the same, though broadening out: stars above, a curve-edged velvet black plain below. They were orbiting.

A pulsing, color-changing glow from somewhere showed him twisted stumps of the radio lattices. There was no sign of the mirror at all. It must

have been torn away, or vaporized completely, in the fiery turbulence of decel.

New maxs showed on the board: Cabin Temperature 214 F, Skin Temperature 907 K, Gravs 87.

Then in the top of the spacefield, almost out of vision, Grunfeld saw the source of the pulsing glow: two sharp-ended ovals flickering brightly all colors against the pale starfields, like two dead fish phosphorescing.



"The torps got to 'em," Croker said, pushed forward beside Grunfeld to the right.

"I did find out at the end," Jackson said quietly from the left, his voice at last free of the trance-tone. "The Enemy ships weren't ships at all. They were (there's no other word for it) space animals. We've always thought life was a prerogative of planets, that space was inorganic. But you can walk miles through the desert or sail leagues through the sea before you notice life and I guess space is the same. Anyway the Enemy was (what else can I call 'em?) space-whales. Inertialess space-whales from the discontinuum. Space-whales that ate hydrogen (that's the only way I know to say it) and spat light to move and fight. The ones I talked to, our brothers, were just their parasites."

"That's crazy," Grunfeld said. "All of it. A child's picture."

"Sure it is," Jackson agreed.

From beyond Jackson, Ness, punching buttons, said, "Quiet."

The radio came on thin and wailing with static: "Titania Station calling fleet. We have jeep and can orbit in to you. The two Enemy are dead—the last in the System. Titania Station calling fleet. We have jeep fueled and set to go—"

Fleet? thought Grunfeld. He turned back to the board. The first and last blue telltales still glowed for *Caliban* and *Starveling*. Breathe a prayer, he thought, for *Moth* and *Snug*.

Something else shone on the board, something Grunfeld knew had to be wrong. Three little words: SHIP ON MANUAL.

The black rim of Uranus ahead suddenly brightened along its length, which was very slightly bowed, like a section of a giant new moon. A bead formed toward the center, brightened, and then all at once the jail-yard sun had risen and was glaring coldly through its pinhole into their eyes.

They looked away from it. Grunfeld turned around.

The austere light showed the captain still in his pressure suit, only the head fallen out forward, hiding the skull features. Studying the monitor box of the captain's suit, Grunfeld saw it was set to inject the captain with power stimulants as soon as the Gravs began to slacken from their max.

He realized who had done the impossible job of piloting them out of Uranus.

But the button on the monitor, that should have glowed blue, was as dark as those of *Moth* and *Snug*.

Grunfeld thought, now he can rest.

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